

myfft

Write your own myfft function identically to the one implemented in the lecture. It takes as input a [Nx1] signal vector **y** and sampling frequency of the Fourier Transform as demonstrated in the lecture, as well as the frequency vector **f**.

The steps are:

- Compute the standard FFT
- Trim away the redundant part
- Compute the frequencies to which each FFT entry corresponds

As an example, the magnitude of a single "impulse" is a vector of all 1's.

```
Fs = 10;
y = [0 0 0 0 1 0 0 0 0 0]';
[yfft, f] = myfft(y,Fs);
```

Your function should give the following result

abs(yfft) is a vector of all 1's and f = (0:5)

Function ?

Save

Reset

 MATLAB Documentation (<https://www.mathworks.com/help/>)

```
1 function [yfft, f] = myfft(y,Fs)
2 %compute the standard fft
3 yfft = fft(y);
4 %trim away the redundant part
5 yfft = yfft(1:ceil((length(y)+1)/2));
6 %compute the frequencies to which each fft entry corresponds.
7 sampnum = [0:(length(yfft)-1)]';
8 f = sampnum * Fs / length(y);
9 end
```

Code to call your function ?

Reset

```
1 % Even length y
2 Fs = 10;
3 y = [0 0 0 0 1 0 0 0 0 0]';
4 [yfft, f] = myfft(y,Fs);
5 abs(yfft')
6
7
8 % Odd length y
9 Fs = 10;
10 y = [0 0 0 0 1 0 0 0 0]';
11 [yfft, f] = myfft(y,Fs);
12 abs(yfft') % All 1's (constant)
13 f' % you should get 0:10/9:40/9
```

Run Function



Assessment: All Tests Passed

Submit



✓ Is output correct with even length random valued inputs?

✓ Is output correct with odd length random valued inputs?

